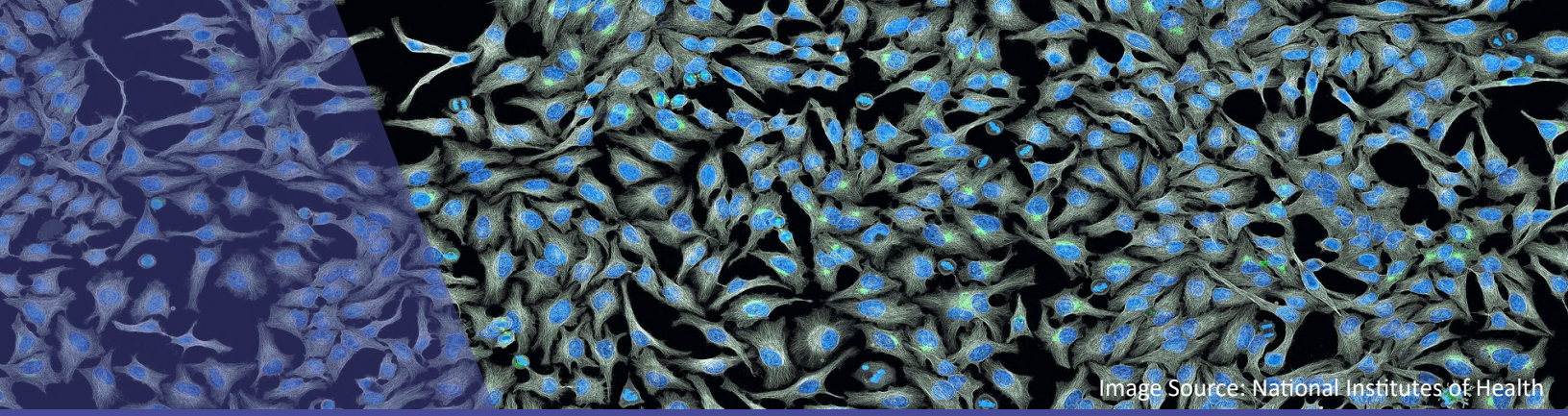


Robert H. Lurie Comprehensive Cancer Center of Northwestern University

Lurie Cancer Center's Basic Research Seminar Series

Decoding How Cells Evolve and Change Fates in Native Human Tissues Via Long-Read Single-Cell RNA Sequencing

Tuesday, November 17, 2026

11:00 a.m.- 12:00 p.m. CT

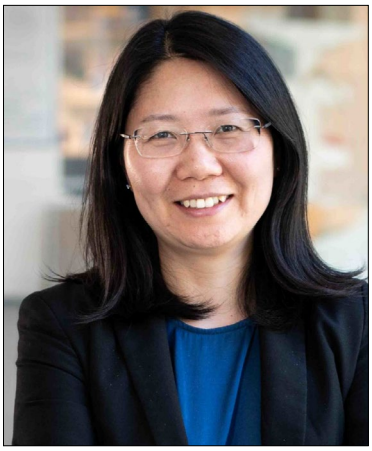
Searle Seminar Room, 1st Floor

Robert H. Lurie Medical Research Center
303 E. Superior St., Chicago, IL

Understanding how cells evolve and change fates in native human tissues requires linking their genetic histories to their transcriptional states. Long-read single-cell RNA sequencing enables joint profiling of expressed genetic variants, gene expression, and transcript isoforms, providing a foundation for resolving these relationships. In this presentation, I will introduce a high-throughput long-read single-cell RNA sequencing approach and computational tools for reconstructing clonal evolutionary lineages from somatic variants and integrating these lineages with transcriptional profiles. Applications to human immune cells and lung tumors reveal the origins and directionality of cell-state changes and the coupling between genetic evolution and fate remodeling. Together, this work establishes a framework for decoding cellular evolution and fate dynamics directly in native tissues without experimental lineage labeling, offering new insights into tumor progression and adaptation in their physiological and disease contexts.

Key topics:
Single-Cell RNA Sequencing and Clonal Evolution

Refreshments will be provided.



Ruli Gao, PhD

Assistant Professor, Biochemistry and Molecular Genetics
Northwestern University Feinberg School of Medicine

Hosted by: Caroline LePoole, PhD

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